

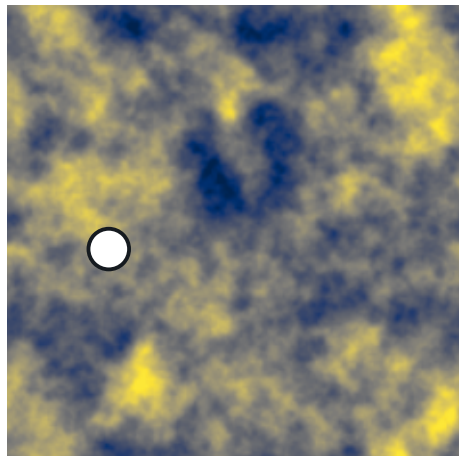
Local adaptation at distance d

how well a population does at home, minus a distance d away

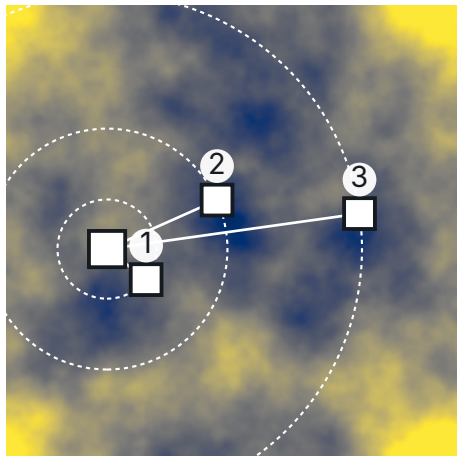
$$\ell_S(d) := \mathbb{E}[\bar{m}_S(0) - \bar{m}_S(d)], \quad S = H, P$$

$\bar{m}_S(d)$: mean growth rate of species S when transplanted a distance d away

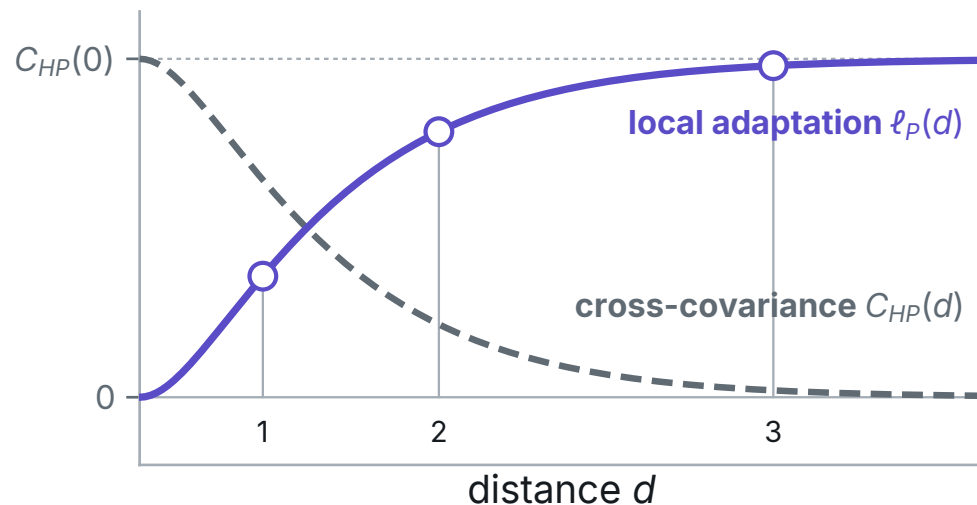
the parasite, at home



hosts: at home, and further and further away



in the model



$$\ell_P(d) = B_P [C_{HP}(0) - C_{HP}(d)]$$

in the model: how well host and parasite match in the same place,
minus how well they match a distance d apart

$$\ell_H(d) = B_H [C_{HP}(d) - C_{HP}(0)]$$

for the host, the mirror image
 B : strength of biotic selection